ENTERPRISE EDITION V2.4.0

ATTENDX ENTERPRISE HR & WORKFORCE MANAGEMENT SYSTEM

Comprehensive Technical Architecture, Geofencing Engine, Multi-Branch Roster, Leave Management, Biometric/QR Gate Kiosk, and Automated Payroll System Manual

DEVELOPED BY :- KALPA WISHVAJITH

CORE WEB FRAMEWORK
Next.js 14 (Pages/API) + TypeScript

DATABASE & ORM
Prisma ORM + PostgreSQL / Supabase

ATTENDANCE & GEOFENCING
Haversine GPS Engine + Dynamic QR Gate

MOBILE ECOSYSTEM
AttendX Mobile (React Native / KMP Ready)

COMPENSATION & FINANCE
Dynamic Payroll Engine + EPF/ETF Deductions

ACCESS CONTROL
5-Tier RBAC + RLS Database Security

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Repository: github.com/midiz-io/HR-Management-System

Document Status: Complete & Verified
Classification: Master Engineering Document

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1. Executive Summary & Core Value Proposition

The **AttendX HR & Workforce Management System** is a cloud-native, enterprise-grade Human Capital Management (HCM) platform engineered to streamline workforce tracking, attendance authentication, smart roster scheduling, multi-tiered leave workflows, overtime governance, employee welfare operations, and automated payroll calculation for modern distributed organizations.

> **Core Architectural Philosophy:**
>
> AttendX bridges physical workplace presence and digital governance using real-time geofencing algorithms, dynamic cryptographic QR gate passes, automated probation/accrual tracking, and complete financial audit trails. It eliminates attendance fraud (buddy punching), automates complex scheduling across geographical branches, and provides unified corporate oversight.

GEOFENCED PUNCHING

Sub-Meter Accuracy

Haversine distance formula validating exact employee coordinates against branch boundaries with dynamic radius checking.

FLEXIBLE ROSTER

Bulk Scheduling

Multi-employee shift scheduling with daily, shift-matching, or custom recurrent day patterns (e.g., Mon/Wed/Sat).

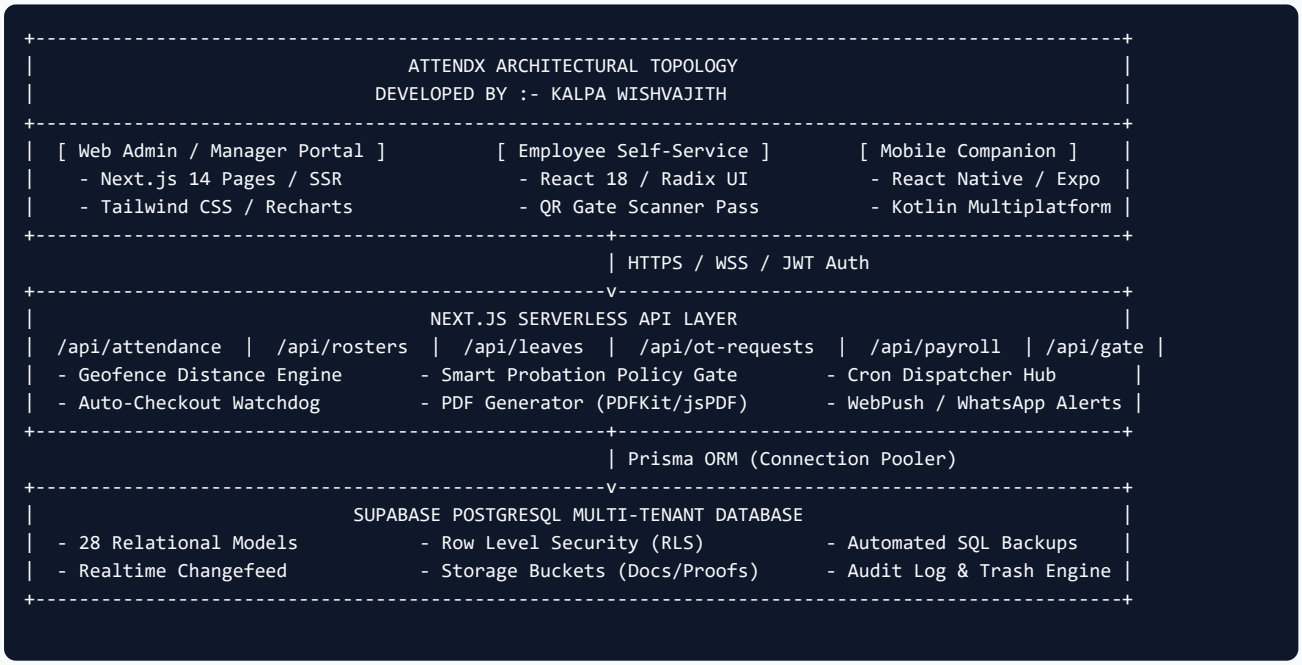
AUTOMATED PAYROLL

Statutory Precision

Integrated salary configurations, overtime calculations, EPF/ETF statutory taxes, salary advances, and digital PDF payslips.

2. System Architecture & High-Level Design

The platform adopts an event-driven, decoupled micro-architecture combining a Next.js serverless backend, real-time database subscriptions via Supabase, client-side caching, and multi-platform mobile consumers.



Backend & Computation Tier

- **Runtime:** Node.js 18+ on Next.js Serverless runtime

Frontend & Mobile Ecosystem

- **UI Framework:** React 18, Radix UI Primitives, Lucide Icons

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- **Database Engine:** PostgreSQL 15 hosted on Supabase
- **Object Relational Mapping:** Prisma 5.19 with transaction safety
- **Connection Pooling:** PgBouncer session & transaction pooling
- **Background Workers:** Cron dispatchers for automated checkouts, late sweeps, and WhatsApp reports

- **Styling System:** Tailwind CSS with responsive layout tokens
- **Visualizations:** Recharts analytical charts & monthly trends
- **Mobile App:** React Native / Expo (`attendx-mobile`) + Kotlin Multiplatform (KMP) master specification
- **Hardware Bridges:** HTML5 QR Scanner, Camera, Geolocation API

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3. Geofencing Attendance & Location Intelligence Engine

The attendance verification subsystem enforces strict geographical validation using the spherical Haversine formula to ensure employees are physically present within authorized office coordinates before a punch is accepted.

Haversine Distance Formulation

The system evaluates coordinates `(lat1, lon1)` from the client device against office boundaries `(lat2, lon2)`:

```
const R = 6371e3; // Earth radius in meters
const phi1 = (lat1 * Math.PI) / 180;
const phi2 = (lat2 * Math.PI) / 180;
const deltaPhi = ((lat2 - lat1) * Math.PI) / 180;
const deltaLambda = ((lon2 - lon1) * Math.PI) / 180;

const a = Math.sin(deltaPhi / 2) * Math.sin(deltaPhi / 2) +
          Math.cos(phi1) * Math.cos(phi2) *
          Math.sin(deltaLambda / 2) * Math.sin(deltaLambda / 2);
const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
const distance = R * c; // Distance in meters

const isWithinGeofence = distance <= allowedRadiusMeters;
```

Attendance Status State Machine

Status	Criteria & Trigger Condition
`PRESENT`	Check-in within shift scheduled window + grace period.
`LATE`	Check-in past grace period; increments Late Tracker.
`ABSENT`	No check-in recorded by scheduled shift end cutoff.
`HALF_DAY`	Work duration meets half-day quota or approved leave.
`SHORT_LEAVE`	Gate pass verified check-out/in within 1.5 - 2 hrs.
`ON_LEAVE`	Active approved leave request covers date.

Automated Checkout & Watchdog

- **Per-User Auto Checkout:** Individual toggle `autoCheckoutEnabled` with customizable working hour thresholds (e.g., 8.0, 9.0 hrs).
- **Cron Auto Checkout:** Scheduled nightly cron job sweeping unclosed check-ins, marking checkout timestamps with an administrative flag.
- **Late Attendance Tracker:** Aggregates consecutive and cumulative late strikes; issues automated warnings and triggers HR review triggers.
- **Kiosk Gate Scanner:** Dedicated physical terminal mode allowing employees to punch via dynamic QR code scanning on security gates.

4. Multi-Shift Scheduling & Bulk Roster Management

Managing shifts across multiple branches and departments is streamlined through the newly developed **Bulk Roster Engine**:

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Scheduling Mode	Behavior & Execution Mechanics	Target Use Case
Daily Repeat	Applies the selected shift to every sequential day across the configured date window without day-of-week skipping.	Continuous operation sites, security officers, 24/7 technical shifts.
Repeat by Shift Days	Filters and matches the default standard working days predefined in the Shift profile (e.g., Monday through Friday).	Headquarters corporate staff, standard operational departments.
Day-by-Day (Custom)	Allows managers to manually toggle active operational days (e.g., Mon, Wed, Sat) with ongoing recurrence or fixed deadlines.	Part-time employees, rotational shift engineers, medical staff.

Off-Day & Shift Swap Governance:

Employees can initiate formal `OffDayRequest` submissions. The workflow enforces a two-tier review hierarchy: the *Department Sectional Head / Recommender* validates team capacity before forwarding to *HR / Reviewer* for final roster mutation.

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5. Leave Governance, Smart Probation & Gate Passes

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The leave management subsystem balances employee flexibility with operational continuity through granular leave type balances and probation protection rules:

Smart Probation Leave Policy

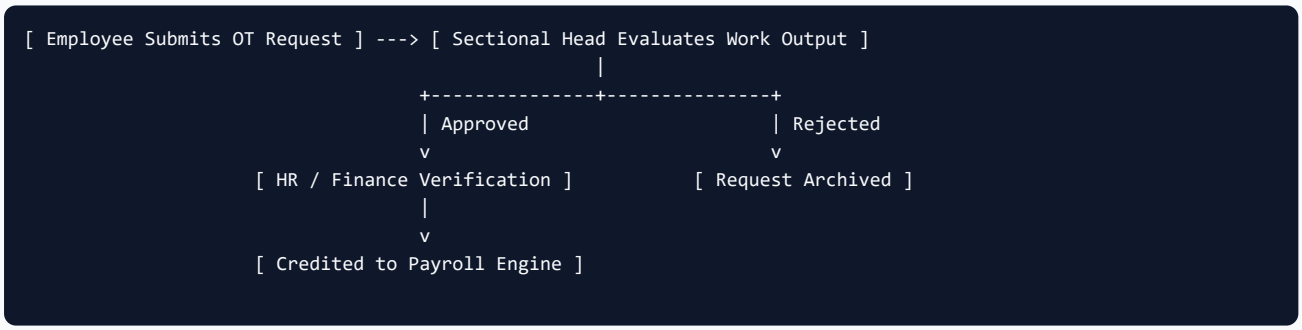
- **3-Month Probation Window:** Calculated automatically from `User.joinDate`.
- **Half-Day Eligibility:** Employees during their initial 90 days are entitled to **one half-day leave per calendar month**.
- **Full-Day Lock:** Full-day annual and casual leaves remain locked until the probation period completes, preventing policy violations.
- **Dynamic Countdown:** Portal displays the exact days remaining until full leave eligibility is unlocked.

Gate Passes & Short Leaves

- **Gate Pass Workflows:** Submissions with official/personal designations, departure time, and vehicle/contact metadata.
- **Security Gate Check:** Dedicated QR Code verification endpoint at security checkpoints records departure and re-entry punches.
- **Short Leave Quota:** Limits short absences (up to 2 hours) to statutory limits without deducting full annual balances.
- **Real-time Status:** Managers monitor pending gate pass checkouts directly on the operational dashboard.

6. Overtime (OT) Automation & Approval Hierarchy

Overtime is managed via structured requests (`OTRequest`) requiring dual validation. Employees request extra hours either proactively or post-shift, accompanied by justification notes and task references.



7. Automated Payroll Engine & Statutory Precision

The payroll module automates salary calculations using employee `SalaryConfig`, incorporating attendance logs, overtime hours, salary advances, statutory taxes, and voluntary corporate deductions.

Component	Classification	Calculation Logic & Source
Basic Salary	Fixed Earnings	Contractual monthly base amount from `SalaryConfig.basicSalary`.
Allowances & Incentives	Variable Earnings	Fixed allowances, attendance bonuses, and performance incentives.
Overtime (OT) Pay	Calculated Earnings	Approved OT hours multiplied by statutory hourly OT multiplier.
Employee EPF (8%)	Statutory Deduction	8% deducted from total earnings eligible for provident fund.

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Component	Classification	Calculation Logic & Source
Employer EPF (12%)	Employer Contribution	12% paid by organization into Central Bank provident scheme.
Employer ETF (3%)	Employer Contribution	3% contributed to Employee Trust Fund.
Salary Advances	Voluntary Deduction	Sum of approved `SalaryAdvanceRequest` payouts for the active cycle.
Welfare / Pending Bills	Internal Deduction	Deductions from seafood orders and employee cafeteria purchases.
Unpaid Absences	Attendance Deduction	Daily rate multiplied by unexcused absent days.

> **Real-Time Daily Salary Banner:**
>
> Employees have access to a real-time daily earnings dashboard component (`EmployeeDailySalaryBanner.tsx`) showing accrued compensation based on verified punches to date, promoting organizational transparency.

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8. Corporate Welfare, Food Orders & Pending Bills

AttendX provides integrated enterprise welfare modules for streamlined cafeteria logistics and staff grocery schemes:

Cafeteria & Meal Ordering

- Daily meal catalog with breakfast, lunch, and dinner selections.
- Kitchen ordering cut-off times with automatic order aggregation.
- HR approval and headcount verification for caterer billing.
- Dietary restrictions and special request tracking.

Seafood Orders & Staff Credit Scheme

- Internal grocery/seafood purchasing with subsidized pricing.
- Sales dispatch status tracking: Ordered -> Dispatched -> Received.
- Dedicated `PendingBillAccount` ledger for staff credit.
- One-click monthly deduction roll-up into the payroll pipeline.

9. Complete Relational Database Architecture (Prisma)

The underlying PostgreSQL database contains 28 robust relational models designed for high-concurrency enterprise workloads:

Entity / Model	Primary Key / Foreign Keys	Purpose & Relationships
`User`	`id (UUID)`, `departmentId`, `locationId`	Central employee record, authentication identity, role, resignation flags, and relation roots.
`Attendance`	`id (UUID)`, `userId`, `locationId`	Daily punch logs, timestamps, coordinates, verification flags, and checkout triggers.
`WorkShift`	`id (UUID)`	Shift templates, start/end schedules, grace durations, and assigned working days.
`Roster`	`id (UUID)`, `userId`, `shiftId`, `locationId`	Calendar schedule assigning an employee to a shift and physical work location.
`LeaveRequest`	`id (UUID)`, `userId`, `reviewerId`	Leave applications, dates, categories, attachments, and hierarchical approval states.
`LeaveBalance`	`id (UUID)`, `userId`, `leaveTypeId`	Available, used, and accrued quotas per employee per leave category.
`OTRequest`	`id (UUID)`, `userId`, `reviewerId`	Overtime hour logging, pre/post justifications, and payroll integration linkage.
`SalaryConfig`	`id (UUID)`, `userId (Unique)`	Base compensation rates, standard allowance matrix, and statutory tax parameters.
`Payroll`	`id (UUID)`, `userId`	Monthly processed salary sheets, net payouts, tax breakdowns, and payslip data.
`GatePassRequest`	`id (UUID)`, `userId`, `reviewerId`	Official/personal exit permissions, vehicle details, and gate QR pass checkout.

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Entity / Model	Primary Key / Foreign Keys	Purpose & Relationships
`CompanyDocument`	`id (UUID)`, `creatorId`	Corporate policies, SOP manuals, and mandatory employee acknowledgement tracking.
`TrashRecord`	`id (UUID)`	Soft-deletion recovery registry enabling point-in-time entity restoration.

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10. REST API Specification & Mobile Contract

The API layer exposes standardized endpoints secured with JWT tokens and Supabase session validation:

Method & Route	Access Role	Description & Payloads
`POST /api/attendance`	Employee / Kiosk	Submit check-in/out punch with latitude, longitude, and device metadata.
`POST /api/rosters/bulk`	Admin / Manager	Assign multi-employee bulk rosters with recurrence patterns (daily, shift, custom).
`POST /api/leaves/request`	Employee	Submit leave with type, dates, reason, and attachment URLs. Validates probation rule.
`GET /api/payroll/process`	Admin / Super Admin	Trigger monthly payroll calculation, EPF/ETF computation, and payslip rendering.
`POST /api/requests/gate-pass`	Employee	Generate exit gate pass. Returns cryptographic token for security gate scanner.
`POST /api/admin/database-backup`	Super Admin	Trigger on-demand SQL dump and sync to cloud redundancy storage.

11. Security, Role-Based Access Control & Database Backup

Role-Based Access Control (RBAC)

- **SUPER_ADMIN:** Unrestricted system control, database backups, financial settings, and user provisioning.
- **ADMIN:** HR management, leave reviews, roster generation, and payroll approvals.
- **MANAGER:** Branch roster adjustments, team attendance auditing, and gate pass approvals.
- **SECTIONAL_HEAD:** Department-level task delegation and off-day recommendations.
- **EMPLOYEE:** Self-service punches, payslip downloads, leave/OT filings, and welfare orders.

Automated Database Backup Engine

- **Automated SQL Snapshots:** Full table schemas, relational data dumps, and foreign key cascades.
- **Dual Redundancy:** Local disk storage + automated Google Drive cloud replication.
- **Point-in-Time Recovery:** Soft-delete trash archive allows instant recovery of deleted users or shifts without database downtime.
- **Secure Vault:** Sensitive keys encrypted at rest using industry-standard protocols.

12. Engineering Ownership & Project Attributions

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Engineered with precision to deliver modern workforce automation, sub-meter geofenced attendance tracking, dynamic multi-shift roster allocation, statutory payroll compliance, and cross-platform mobile connectivity.

Official Repository: https://github.com/midiz-io/HR-Management-System • All Rights Reserved
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